

H.S. 2nd Year — Computer Science and Applications (2023)

Solved Practical Question Paper — Code: 33T CSCA (PQ) (Bengali)

Full Marks: 30 | Pass Marks: 12 | Time: Three hours

Q. No. 1 — Answer any one (10 marks) [All five options solved]

(a) Write a C++ program to find the largest element of an array of size 12.

```
#include <iostream>
using namespace std;
int main()
{
    int a[12], i, largest;
    cout << "Enter 12 elements:\n";
    for (i = 0; i < 12; i++)
        cin >> a[i];
    largest = a[0];
    for (i = 1; i < 12; i++)
    {
        if (a[i] > largest)
            largest = a[i];
    }
    cout << "\nLargest element = " << largest << endl;
    return 0;
}
```

Sample Output:

```
Enter 12 elements:
4 9 2 45 12 7 33 21 8 19 3 50
Largest element = 50
```

(b) Write a C++ program to sort the elements of an array using insertion sort.

```
#include <iostream>
using namespace std;
int main()
{
    int n, i, j, key;
    cout << "Enter number of elements: "; cin >> n;
    int a[n];
    cout << "Enter " << n << " elements:\n";
    for (i = 0; i < n; i++)
        cin >> a[i];
    for (i = 1; i < n; i++)
    {
        key = a[i];
        j = i - 1;
        while (j >= 0 && a[j] > key)
        {
            a[j + 1] = a[j];
            j--;
        }
        a[j + 1] = key;
    }
    cout << "\nSorted array:\n";
}
```

```

        for (i = 0; i < n; i++)
            cout << a[i] << " ";
        return 0;
}

```

Sample Output:

```

Enter number of elements: 5
Enter 5 elements:
9 3 7 1 5
Sorted array:
1 3 5 7 9

```

(c) Write a C++ program to implement PUSH() and POP() operations on a stack.

```

#include <iostream>
#define MAX 5
using namespace std;
int stack1[MAX], top = -1;
void push(int val)
{
    if (top == MAX - 1)
        cout << "Stack Overflow!" << endl;
    else
    {
        top++;
        stack1[top] = val;
        cout << val << " pushed onto stack." << endl;
    }
}
void pop()
{
    if (top == -1)
        cout << "Stack Underflow!" << endl;
    else
    {
        cout << stack1[top] << " popped from stack." << endl;
        top--;
    }
}
int main()
{
    push(10);
    push(20);
    push(30);
    pop();
    push(40);
    pop();
    pop();
    return 0;
}

```

Sample Output:

```

10 pushed onto stack.
20 pushed onto stack.
30 pushed onto stack.
30 popped from stack.

```

40 pushed onto stack.
40 popped from stack.
20 popped from stack.

(d) Write a C++ program for creation and displaying a binary file.

```
#include <iostream>
#include <fstream>
using namespace std;
int main()
{
    int num;
    // Writing to binary file
    ofstream outFile("data.dat", ios::binary);
    cout << "Enter 5 numbers to store in binary file:\n";
    for (int i = 0; i < 5; i++)
    {
        cin >> num;
        outFile.write((char*)&num, sizeof(num));
    }
    outFile.close();
    // Reading and displaying from binary file
    ifstream inFile("data.dat", ios::binary);
    cout << "\nData read from binary file:\n";
    while (inFile.read((char*)&num, sizeof(num)))
    {
        cout << num << " ";
    }
    inFile.close();
    return 0;
}
```

Sample Output:

Enter 5 numbers to store in binary file:
10 20 30 40 50
Data read from binary file:
10 20 30 40 50

(e) Write a C++ program to add two matrices of order 3×3.

```
#include <iostream>
using namespace std;
int main()
{
    int a[3][3], b[3][3], c[3][3], i, j;
    cout << "Enter elements of first 3x3 matrix:\n";
    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            cin >> a[i][j];
    cout << "Enter elements of second 3x3 matrix:\n";
    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            cin >> b[i][j];
    for (i = 0; i < 3; i++)
        for (j = 0; j < 3; j++)
            c[i][j] = a[i][j] + b[i][j];
    cout << "\nSum of the two matrices:\n";
}
```

```

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
            cout << c[i][j] << "\t";
        cout << endl;
    }
    return 0;
}

```

Q. No. 2 — SQL from Student table (any three) [All five solved, 1×3=3]

Table structure used below: Student (Student_ID, S_name, Marks)

(a) Write SQL to display all the S_name and Marks.

```
SELECT S_name, Marks FROM Student;
```

(b) Write SQL to find the S_name who has got marks less than 45.

```
SELECT S_name FROM Student WHERE Marks < 45;
```

(This will display Jotin and Rekha.)

(c) Write SQL to display all the data in ascending order of marks.

```
SELECT * FROM Student ORDER BY Marks ASC;
```

(d) Write SQL to find the maximum marks from the student table.

```
SELECT MAX(Marks) FROM Student;
```

(This will display 70, the marks of Mary.)

(e) Write SQL to find the total no. of records available in student table.

```
SELECT COUNT(*) FROM Student;
```

(This will display 5.)

Q. No. 3 — Project Work (6 marks)

Ans: The Project Work is an individual practical assignment developed by the student using C++/SQL, evaluated by the examiner. It cannot be pre-'solved' generically. Suggested guidance: choose a simple real-life problem (e.g., Student Management, Library, Billing System); include title, objective, source code, sample input/output and conclusion; use OOP concepts and, if a database project, proper SQL commands; keep it neatly documented and be ready to explain the logic during viva.

Q. No. 4 — Practical File (6 marks)

Ans: The Practical File is the record maintained by the student throughout the year containing all programs performed in the lab (C++ programs and SQL queries), each with aim, code and output. It is evaluated on completeness, correctness, and neatness rather than being a fixed question. Ensure it includes all programs done in class, is page-numbered, and is verified/signed by the subject teacher before the exam.

Q. No. 5 — Viva Voce (5 marks)

Ans: Viva voce is an oral question-answer session based on the practical syllabus (C++ programming, OOP, data structures, DBMS/SQL, networking). Be thorough with basic theory concepts, be able to explain your own project

and every program in your practical file line by line, and revise short-answer topics, since viva questions usually mirror these.